

Deep Site-Occupancy Models: Scaling Species Trend Estimation from Citizen Science Data

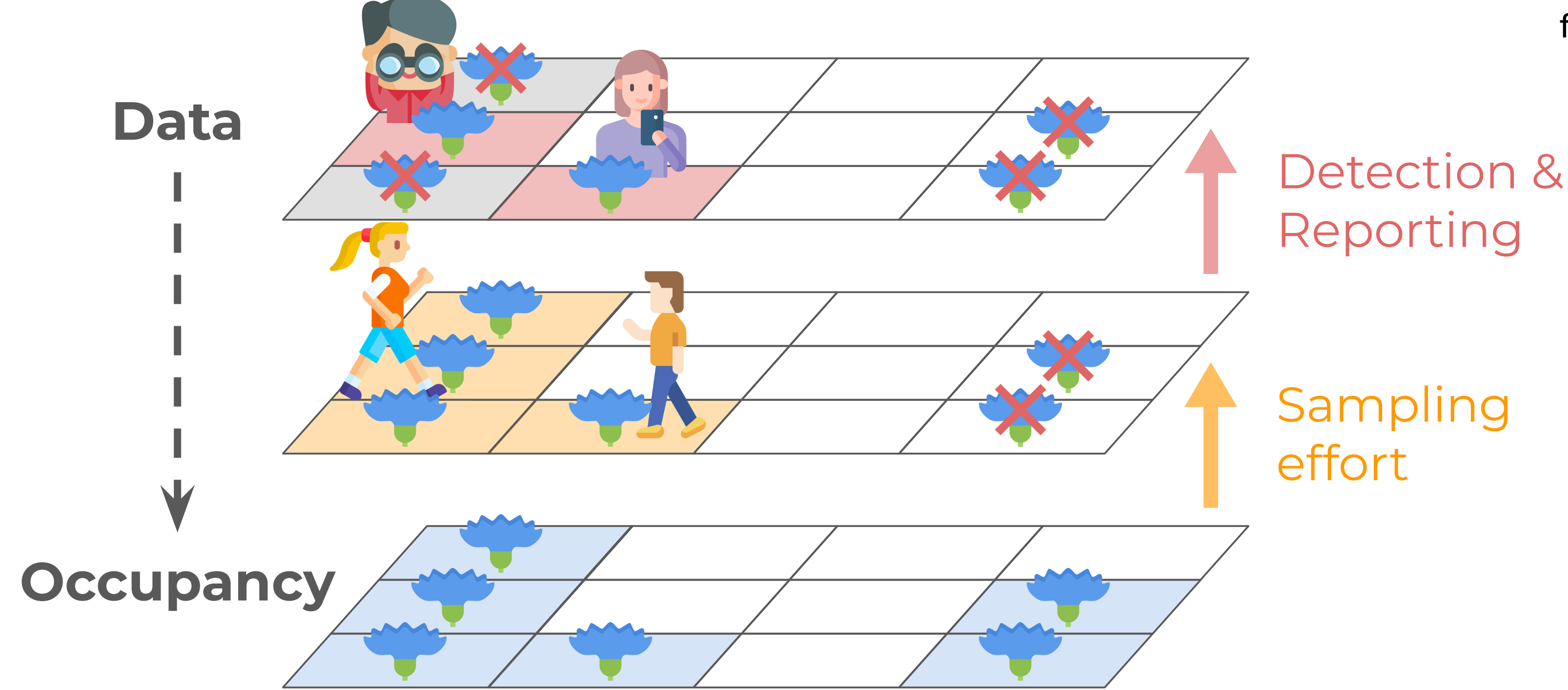
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Opportunistic data

Presence-only opportunistic data from citizen science projects offer unprecedented opportunity to monitor biodiversity, particularly **trends** and **species distributions**.

However, they are affected by **biases**, such as sampling bias and observer biases (e.g. detection and reporting).

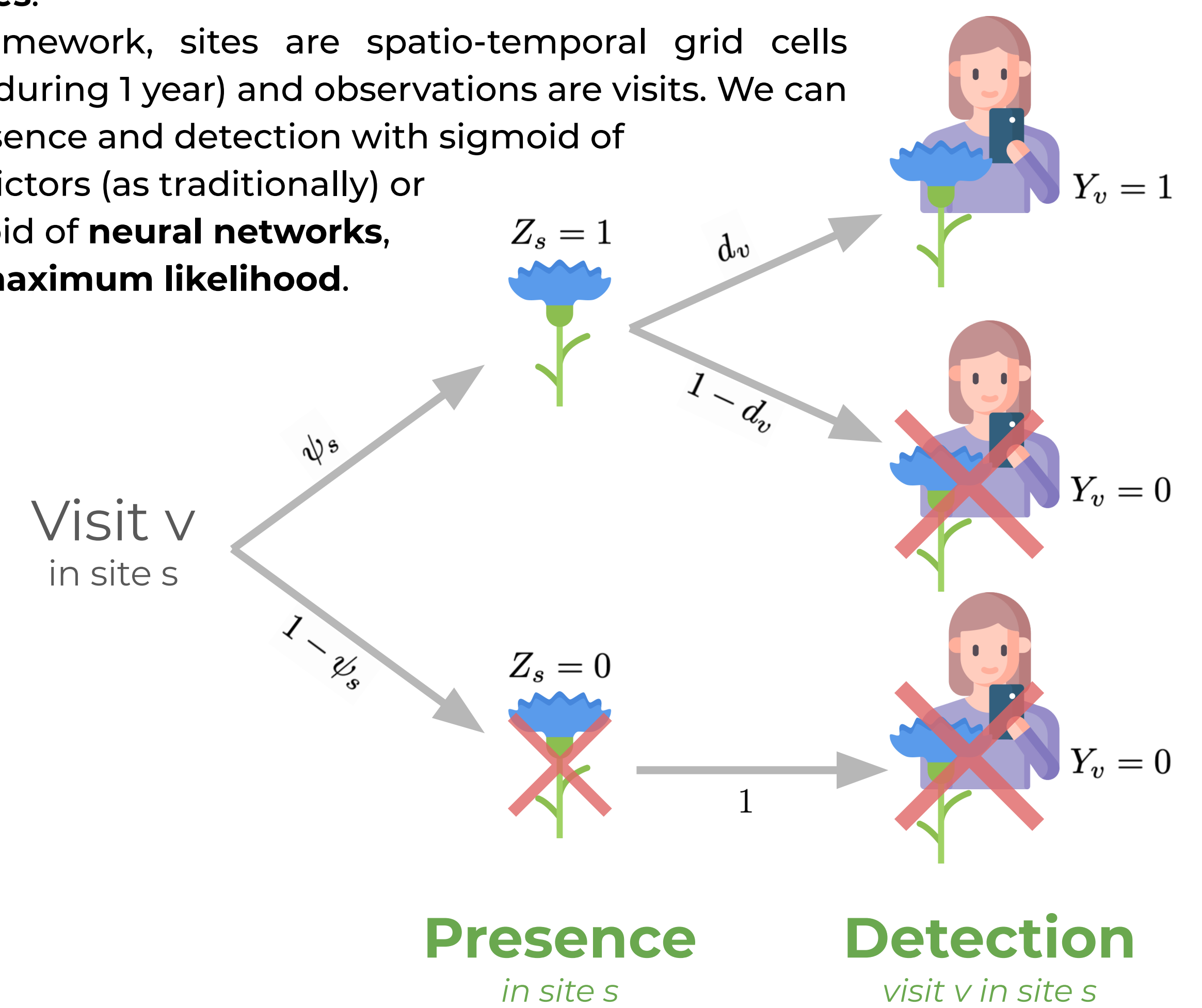
Addressing such biases could allow us to better **monitor and study biodiversity**, especially in the current context of **global changes** and **biodiversity crisis**.



Site occupancy models

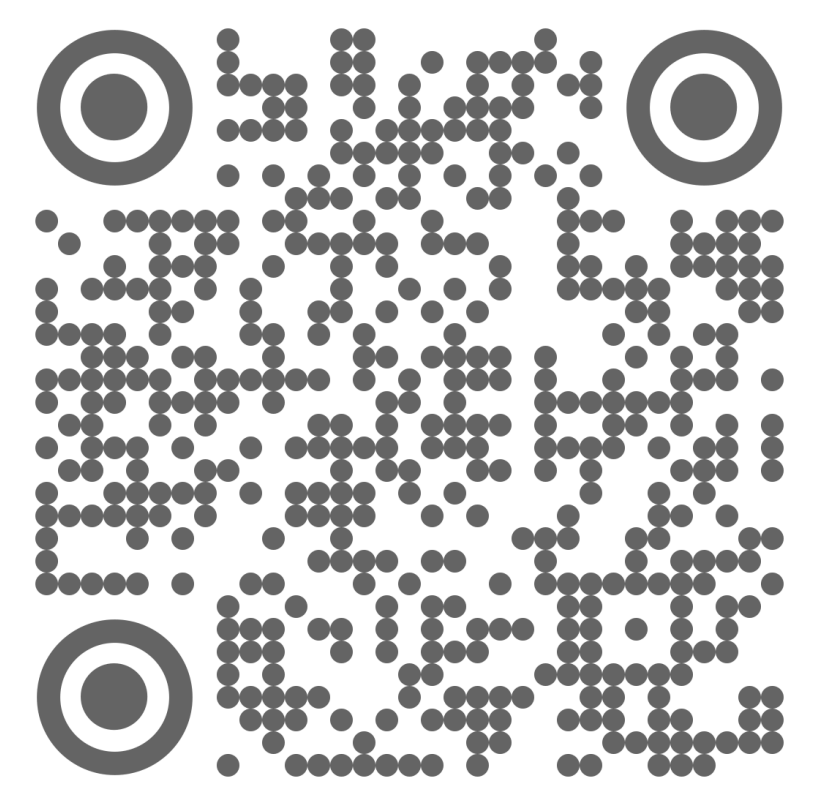
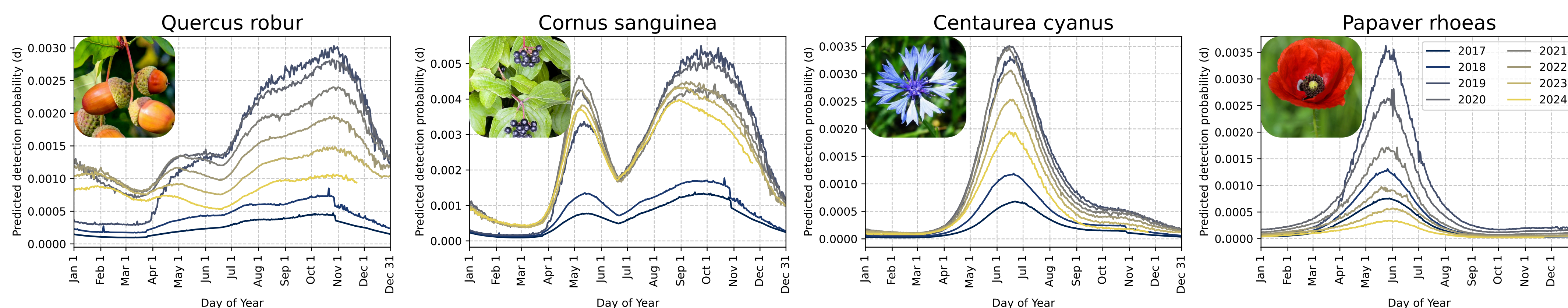
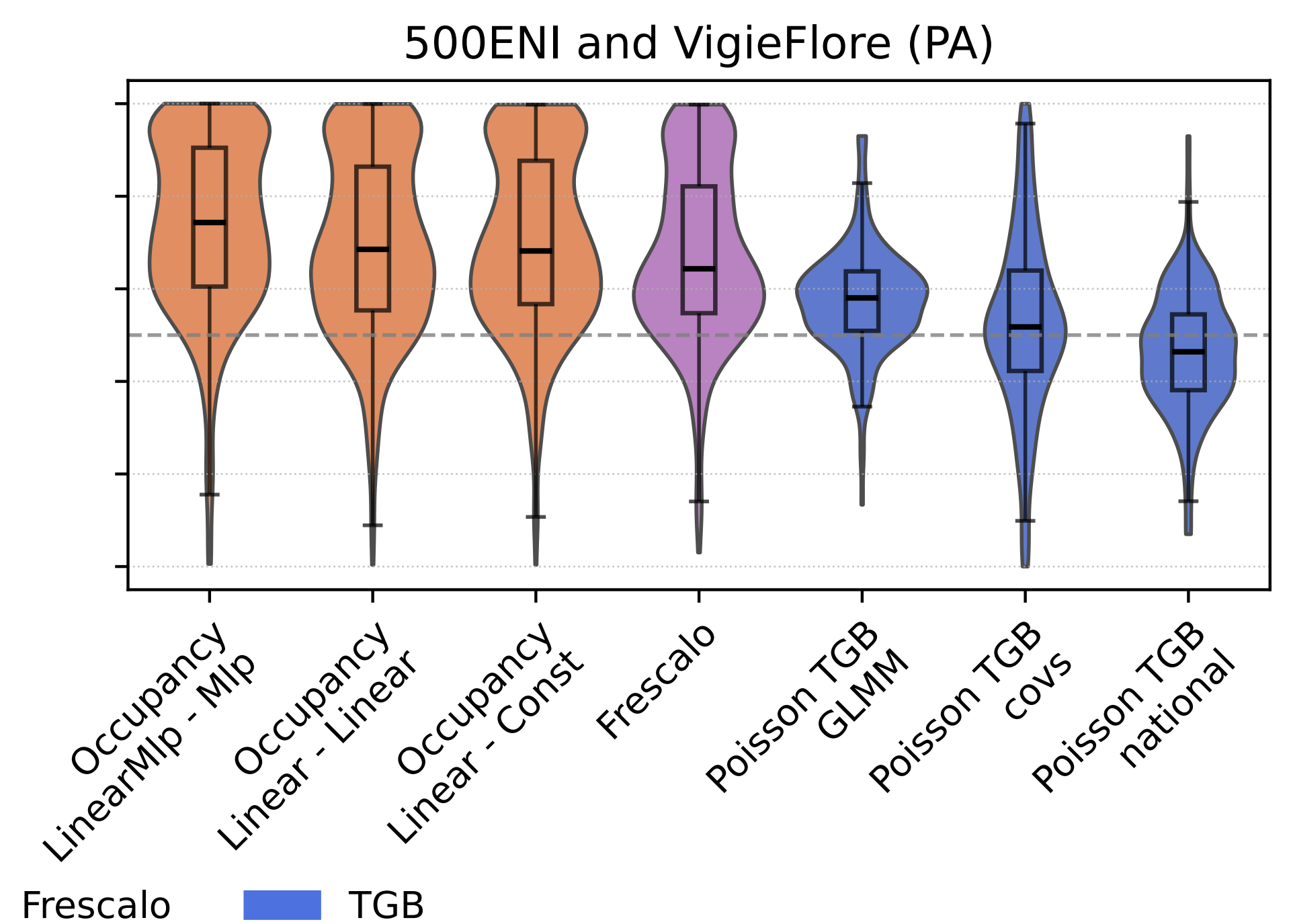
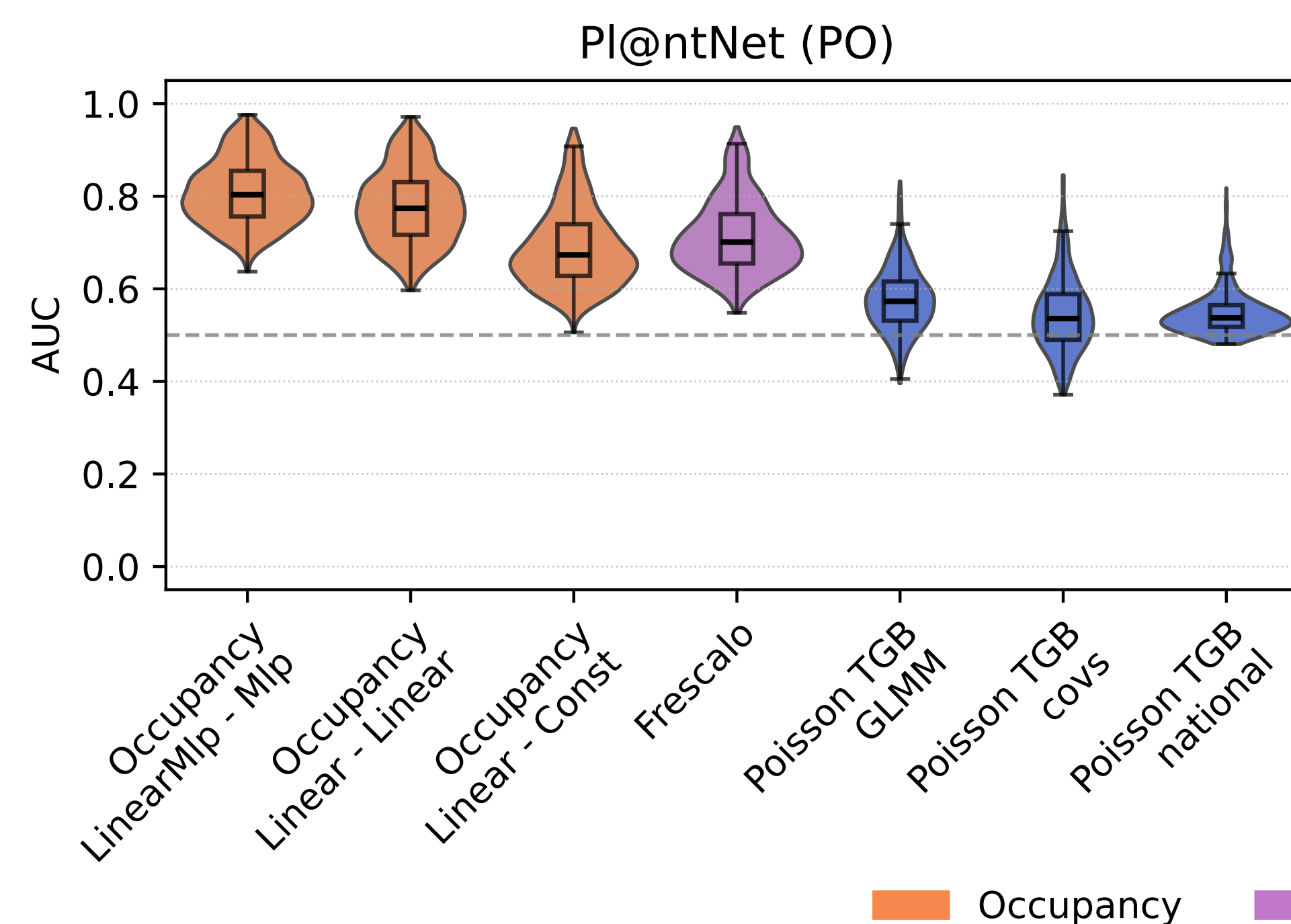
Site occupancy models allow to **correct both sampling effort and detection biases to estimate presence probabilities**.

In our framework, sites are spatio-temporal grid cells (25×25 km during 1 year) and observations are visits. We can model presence and detection with sigmoid of linear predictors (as traditionally) or with sigmoid of **neural networks**, fitted via **maximum likelihood**.



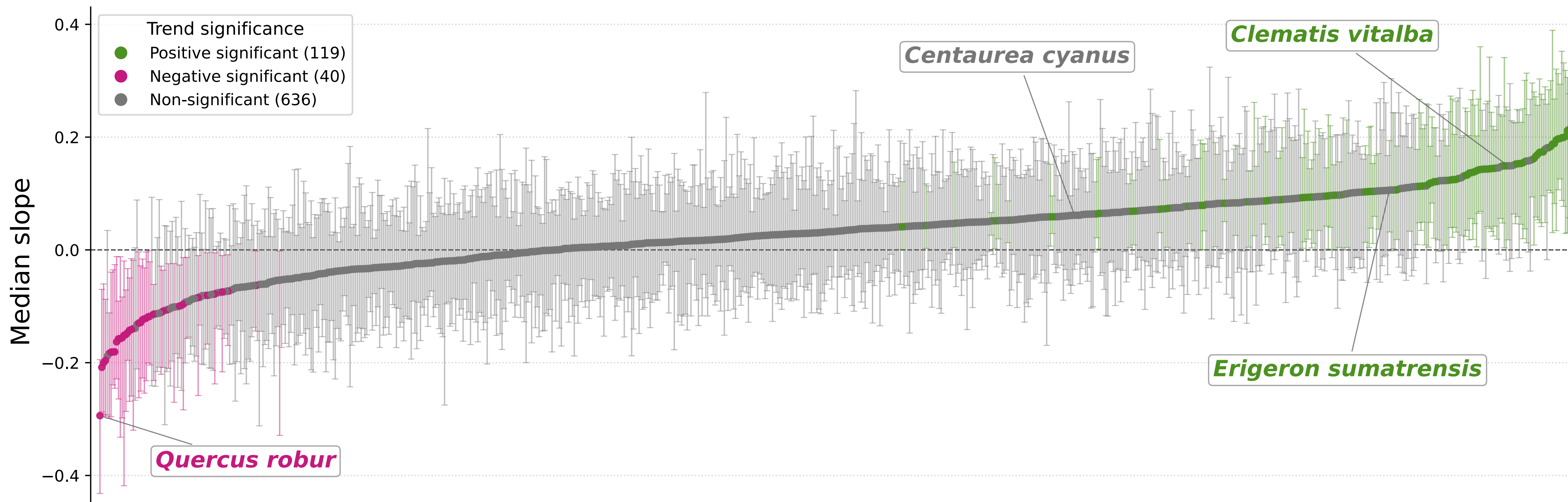
Key Results

- **Occupancy models fit better:** higher AUC on both presence-only and independent presence-absence datasets compared to traditional methods.
- **Expressivity matters:** Using neural networks provides greater expressivity and captures more signal from the data. Nonlinear patterns in both presence and detection emerge, such as peaks of detection related to phenology (e.g. fruiting or flowering).
- **Occupancy trends reveal detection bias correction:** All methods show positive correlation in occupancy trend estimates ($r = 0.29-0.95$). However, expressive occupancy models diverge more from traditional methods than simpler models, suggesting that better detection bias modeling uncovers different ecological trend signals in species occupancy.



Outlook

- A **Python package** for fitting occupancy models on large, uneven datasets, supporting any PyTorch-compatible model for both occupancy and detection components (including neural networks).
- Apply these frameworks to **other datasets and occupancy-model variants**. This will help leverage the vast amount of opportunistic biodiversity data available for species distribution modelling across space and time.
- **Improve occupancy and detection modelling** by exploring more advanced architectures (e.g. CNNs, attention mechanisms) and additional data sources (e.g. satellite imagery, observer histories).



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References

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